Question	Answer	Marks	Partial Marks
5(d)(ii)	48.75 nfww	**4**	**M1** for midpoints soi **M1** for use of $\sum fx$ with *their* frequencies **M1** (dep on 2nd M1) for $\sum fx \div$ (60 or by *their* $\sum f$)
6(a)(i)	Angle *ABC*=52 nfww	**B1**	ALTERNATIVE [Reflex] angle *AOC* = 256
	Opposite angles in cyclic quad oe Angles in opposite segments	**B1**	Angle at centre=2 × angle at circumference/arc
	[Angle *AOC*=104] Angle at centre=2 × angle at circumference/arc nfww	**B1**	Angles around a point
6(a)(ii)	22 nfww	**2**	**B1** for angle *OAC* = 38 or angle *CAD* = 24
6(a)(iii)	28	**1**	
6(a)(iv)	36.6 or 36.62 to 36.63 nfww	**3**	**B2** for 7.4 or 17.42 to 17.43 or **M2** for $9.6\times 2+\frac{104}{360}\times 2\times\pi\times 9.6$ or **M1** for $\frac{104}{360}\times 2\times\pi\times 9.6$
6(b)(i)	81	**3**	**M2** for $\frac{A}{36}=\left(\sqrt[3]{\frac{2187}{648}}\right)$ oe or better or for $A\times\frac{648}{36}\times\sqrt[3]{\frac{2187}{648}}=2187$ oe or better or **M1** for $\frac{A^3}{36^3}=\frac{2187^2}{648^2}$ oe or $\sqrt[3]{\frac{2187}{648}}$ or $\sqrt[3]{\frac{648}{2187}}$
6(b)(ii)	8.05 or 8.051 to 8.052…	**3**	**M2** for $\left[r^3=\right]\frac{2187\times 3}{4\times\pi}$ oe or **M1** for $\frac{4\pi r^3}{3}=2187$ **SC2** for $\frac{648\times 3}{4\times\pi}$ or **SC1** for $\frac{4\pi r^3}{3}=648$
7(a)	Reflection $y=-1$	**2**	**B1** for each
7(b)(i)	Image at (–6, 5) (–6, 7) (–5, 7) (–4, 5)	**2**	**B1** for translation by $\begin{pmatrix}-3\\k\end{pmatrix}$ or $\begin{pmatrix}k\\4\end{pmatrix}$

Question	Answer	Marks	Partial Marks
7(b)(ii)	Image at (1, –1) (3, –1) (3, –3) (2, –3)	**2**	**B1** for shape correct size and orientation but wrong position
7(b)(iii)	Image at (1, 2) (1, 6) (3, 6) (5, 2)	**2**	**B1** for shape correct size and orientation, wrong position
8(a)(i)	$\frac{2}{5}$ oe	**2**	**M1** for $\frac{4}{6}\times\frac{3}{5}$
8(a)(ii)	$\frac{3}{5}$ oe	**1**	**FT** 1 – *their* $\frac{12}{30}$ oe
8(b)	$\frac{5}{7}$ oe nfww	**4**	**M3** for $\frac{2}{7}+\frac{5}{7}\times\frac{2}{6}+\frac{5}{7}\times\frac{4}{6}\times\frac{2}{5}$ oe or for $1-\frac{5}{7}\times\frac{4}{6}\times\frac{3}{5}$ oe or **M1** for each of $\frac{5}{7}\times\frac{2}{6}$ and $\frac{5}{7}\times\frac{4}{6}\times\frac{2}{5}$ oe or completed tree diagram with appropriate probabilities shown
9(a)(i)	5	**1**	
9(a)(ii)	1	**2**	**M1** for h(0) or 3^{9-x^2} or better
9(a)(iii)	$9 – 4x^2$ final answer	**1**	
9(a)(iv)	$15 – 2x^2$ final answer	**2**	**M1** for $2(9 – x^2) –3$ or better
9(b)	$\frac{x+3}{2}$ final answer	**2**	**M1** for $x = 2y – 3$ or $y + 3 = 2x$ or better or $\frac{y}{2}=x-\frac{3}{2}$
9(c)	1.8 or $1\frac{4}{5}$ or $\frac{9}{5}$	**2**	**M1** for $10x-15=3$ or $2x-3=\frac{3}{5}$
9(d)	–1 and 4 nfww	**4**	**M1** for $9 – (2x – 3)^2= –16$ **A1** for $4x^2 – 12x – 16[= 0]$ oe **M1** (dep on first M1) for correct factors or use of formula or completing the square for their 3-term quadratic OR **M1** for $9 – y^2= –16$ **A1** for $y^2 = 25$ **M1** (dep on first M1) for $2x – 3= \pm5$
9(e)	$\frac{1}{9}$	**1**	